O-S-576F
August 3, 1974
SUPERSEDING
Fed. Spec. O-S-576E
April 3, 1967

FEDERAL SPECIFICATION

SODIUM BICARBONATE, TECHNICAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for one type and two classes f sodium bicarbonate ($NaHCO_3$) (see 6.1).

1.2 Classification. The sodium bicarbonate covered by this specification shall be f the following classes, as specified (see 6.2).

Class 1 - Coarse powder (see 3.1.1).
Class 2 - Powdered (see 3.1.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids r request for proposal, form a part of this specification to the extent specified erein.

eral Specifications:

QQ-S-781 - Strapping, Steel, Flat and Seals.
RR-S-366 - Sieve, Test.
UU-S-48 - Sacks, Shipping, Paper.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood, Cleated-plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-F-320 - Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-001660 - Pallet, Expendable.
PPP-S-760 - Strapping, Nonmetallic (and Connectors).

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activitites outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of ederal Specifications and Standards at the prices indicated in the Index. The Index, hich includes cumulative monthly supplements as issued, is for sale on a subscription asis by the Superintendent of Documents, U.S. Government Printing Office, Washington, 20402.

(Single copies of this specification and other Federal Specifications required by tivities outside the Federal Government for bidding purposes are available without harge from Business Service Centers at the General Services Administration Regional fices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, rt Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, andards, and Handbooks and the Index of Federal Specifications and Standards from tablished distribution points in their agencies.)

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Military Specification:

MIL-B-117 - Bag, Sleeve and Tubing - Interior Packaging.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Form. The sodium bicarbonate shall be in powder form. Any caking shall be capable of being readily disintegrated.

3.1.1 Particle size. The particle size of classes 1 and 2 shall conform to table I, when tested in accordance with 4.2.2.

TABLE I. Particle Size

Requirement	Percentage			
	Class 1		Class 2	
	Min.	Max.	Min.	Max.
Pass through No. 100 sieve	99.0	-	99.0	-
Pass thorugh No. 200 sieve	50.0	70.0	55.0	80.0

3.2 Insoluble matter. The matter insoluble in water shall be not more than 0.1 percent by weight when tested in accordance with 4.2.2.

3.3 Loss on heating. The loss on heating shall be not less than 35.5 percent when tested in accordance with 4.2.2.

3.4 Chloride. The sodium bicarbonate shall pass the test described in 4.2.2.

3.5 Total alkalinity. The total alkalinity of the material shall be not less than 98.0 nor more than 100.3 percent when determined in accordance with the requirements of 4.2.2.

3.6 Moisture. The free moisture shall be not greater than 0.3 percent when tested in accordance with 4.2.2.

3.7 Workmanship. The product shall be clean, uniform, and free from any defects which impair its utility.

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QUALITY ASSURANCE

Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Sampling for inspection shall be in accordance with MIL-STD-105 except where otherwise indicated.

4.2 Inspection of the end item.

4.2.1 Examination of the end item. The end item shall be examined for the defects in applicable sub-paragraphs below and at the inspection levels and acceptable quality levels (AQLs) set forth in 4.2.1.5. The lot shall be expressed in units of filled unit containers of the same capacity for examination in 4.2.1.1, 4.2.1.2, 4.2.1.3 and in units of shipping containers for examination for preparation for delivery under 4.2.1.4. A lot shall consist of all sodium bicarbonate offered for inspection at one time which has been produced by one manufacturer under essentially the same manufacturing conditions and with no change in raw materials and made in a continuous manufacturing process. In the event that the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.1.1 Examination of the unit container. The sample unit for this examination shall be one filled unit container.

Examine	Defect
Construction	Not as specified.
Workmanship	Any seepage.
	Any breaks.
	Dirty.
Closure	Not as specified.
	Incomplete, imperfect.
Markings	Missing, incomplete, incorrect.
	Not clear.
	Not legible.

4.2.1.2 Examination of contents of the unit container. The sample unit for this examination shall be the contents of one filled unit container.

Examine	Defect
Appearance	Presence of foreign material.
	Not uniform.
	Contains agglomerates, cakes or lumps which cannot be readily broken up by hand.
	Not clean.
	Not powder.

4.2.1.3 Examination of the end items for net contents. The sample unit for this examination shall be one filled container. The lot shall be unacceptable if the average net contents per container for all sample units examined is less than specified or indicated.

4.2.1.4 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, marking and unit loads (palletization) comply with the requirements of section 5. Defects shall be scored as specified in table II. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

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TABLE II. Examination of preparation for delivery

Examine	Defect
Containers	Not as specified.
Contents	Not as specified.
Markings	Omitted: incorrect; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers.
Palletization, when specified	Not as specified.

4.2.1.4.1 Examination of closure and reinforcement of containers. When shipping containers are required to comply with UU-S-48 or PPP-B-636, the examination for defects in closure and reinforcement shall be in accordance with appendix of the same specification.

4.2.1.5 Inspection levels and acceptable quality levels (AQLs) for examination. The inspection levels for determining the sample size, and the acceptable quality levels (AQLs) expressed in defects per one hundred units shall be as follows:

Examination paragraph	Inspection level	AQLs
4.2.1.1	S-3	2.5
4.2.1.2	S-3	2.5
4.2.1.3	S-3	-

4.2.2 Testing of the end item. The end item shall be tested for the characteristics listed in table III for each lot. The lot size shall be expressed in pounds of sodium bicarbonate. The sample shall be a twelve-ounce composite selected in equal portions from samples selected at random in the frequency listed below and thoroughly mixed to yield the composite. The sample shall be placed in an airtight, watertight, glass container with suitable caps, covered, sealed to prevent any moisture change and labeled to identify the lot and container from which it was taken. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if the composite sample fails to meet any test requirement specified.

Lot size (pounds)	Sample frequency
800 or less	2
801 up to and including 22,000	3
22,001 or more	5

4.3 Test methods.

4.3.1 Particle size. The sieves used (see 3.1.1) shall comply with RR-S-366, with the No. 100 sieve on the top and then the No. 200 sieve on the bottom pan. Place 100 grams of the sample on the upper sieve, cover, and shake for ten minutes by means of a mechnical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the amounts of material retained by the sieves and calculate the percentage retained on each sieve. In humid weather, before testing, heat the sieves and sample in an oven at maximum temperature of 45°C. for ten to fifteen minutes to prevent sieve binding.

4.3.2 Insoluble matter. Dissolve 20 g of the sodium bicarbonate, weighed to the nearest 0.001 g, in 300 ml of water at room temperature. Filter through a previously prepared, dried, and weighed Gooch or fritted glass crucible. Wash the residue free of alkali with water and dry the crucible in an oven at 100° ± 5°C.

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TABLE III. Instructions for testing of the end item

acteristic	Specification reference Require-ment	Test method	Requirements applicable to Indiv. unit	Comp. samp.	Number deter-minations per unit	Results reported as Pass or Fail	Numerically to nearest
acteristic 9 micron o. 100) eve	3.1.1	4.3.1		X	Average of 2		1.0 percent
icron o. 200) eve	3.1.1	4.3.1		X	Average of 2		1.0 percent
luble matter	3.2	4.3.2		X	1	X	
on heating, cent	3.3	4.3.3		X	Average of 2		0.1 percent
rides	3.4	4.3.4		X	1	X	
l alinity	3.5	4.3.5		X	Average of 2		1.0 percent
ure	3.6	4.3.6		X	Average of 2		0.01 percent

4.3.2.1 Calculation. Calculate the percentage of matter insoluble in water as ows:

$$\text{Matter insoluble in water, percent} = \frac{\text{Wt. of residue to nearest 0.01 g} \times 100}{\text{Wt. of } NaHCO_3 \text{ to nearest 0.01g}}$$

4 Loss on heating. Dry about 4 g of the specimen in a desiccator over con-rated sulfuric acid at room temperature (20° to 30°C.) for 24 hours. Weigh the l specimen to the nearest milligram (mg) in a tared porcelain crucible. Ignite in fle furnace at a temperature of 275° ± 5°C. to constant weight (about 3 hours).

4.3.3.1 Calculation. Calcu te the percent loss in weight as follows:

$$\text{Percent loss on heating} = \frac{100\ (W-A)}{W}$$

: A = g of residue after heating.
W = g of specimen before heating.

4.3.4 Chloride. Dissolve 5 g of the specimen, weighed to the nearest 0.01 g in l of water. Transfer to a 1-liter volumetric flask and bring to volume. To 10 this solution in a test tube, add 1 ml of a 1 to 10 solution of nitric acid and and 1 ml of 0.1 N silver nitrate. The turbidity shall not be greater than that ced by 0.03 mg of sodium chloride in 10 ml of water to which are added the same ities of acid and silver nitrate as used with the specimen.

4.3.5 Total alkalinity. Weigh 2 or 3 g of the material, as received, to the st mg. Dissolve this in 50 ml of water and titrate with 1 N sulfuric acid using l orange as indicator.

4.3.5.1 Calculation. Calculate the total alkalinity as sodium bicarbonate as ws:

$$\text{Total alkalinity (as sodium bicarbonate)} = \frac{8.4\ VN}{W}$$

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where:

V = ml of sulfuric acid solution used.
N = normality of sulfuric acid solution.
W = g of sodium bicarbonate used.

4.3.6 Moisture. Place 10 g of the sodium bicarbonate in a weighed crucible and weigh to ±0.01g. Dry this sodium bicarbonate and crucible in a sulfuric acid desiccator for four (4) hours. Reweigh the sample.

4.3.6.1 Calculation. Calculate the percentage of moisture as follows:

$$\text{Percent moisture} = \frac{\text{Loss in weight in grams} \times 100}{\text{Weight of sample in grams}}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. One pound or 1-1/2 pounds of sodium bicarbonate as specified (see 6.2), shall be packaged in a sift-proof paperboard box commercially used for the specific product being packaged.

5.1.2 Level C. The sodium bicarbonate shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 One pound and one-half pound quantities. Forty-eight one pound or forty-eight 1-1/2 pound boxes of sodium bicarbonate, packaged as specified in 5.1 shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636, or cleated-plywood box conforming to overseas type, style optional, grade A preserved of PPP-B-601, or wirebound-wood box conforming to class 3, style optional, of PPP-B-565. Each fiberboard shipping container shall be closed and waterproofed by means of tape as specified in the appendix of PPP-B-636. Each cleated-plywood and wirebound box shall be furnished with a waterproof bag liner formed from material meeting the minimum requirements of type II, class B, style 1 or 2 of MIL-B-117, except that asphalt laminated material shall not be permitted. Inside faces of each wood box shall be lined with fiberboard meeting minimum requirements of grade W6c of PPP-F-320. Each wood box shall be closed as specified in the appendix to the applicable box specification. Also, each box shall be reinforced by banding or strapping as specified in the appendix to the applicable box specification.

5.2.1.2 One hundred pound quantity. One hundred pounds of sodium bicarbonate shall be packed in a paper shipping sack conforming to construction no. 15 or 15 X, type and class optional, style B, grade MB3 protection of UU-S-48, except that asphalt shall not be used in the moisture barrier (MB) material. Valve sack closure, if used, shall have sleeves tucked in. Sleeves shall be made from moisture-proof paper having an inside lining of polyethylene. After the sacks have been filled, the sleeves shall be heat-sealed or closed by tape. Ends of the sack shall be heat-sealed after stitching, and secured with tape in accordance with the sack specification. As an alternate to heat sealing, ends of the sack may be sealed by dipping in accordance with the sack specification.

5.2.2 Level B.

5.2.2.1 One pound and one and one-half pound quantities. Unless otherwise specified (see 5.2.2.1.1), forty-eight 1-pound or forty-eight 1-1/2 pound boxes of sodium bicarbonate, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to class domestic of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of the container specification.

5.2.2.1.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636, and closed in accordance with the appendix of the container specification.

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5.2.2.2 One hundred pound quantity. One hundred pounds of sodium bicarbonate shall be packed in a paper shipping sack conforming to construction of UU-S-48.

5.2.3 Level C. Sodium bicarbonate in quantities as specified (see 6.2) packaged as specified in 5.1 shall be packed in containers to assure carrier acceptance and safe arrival at destination at the lowest transportation rate for such supplies. Containers shall be in compliance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.3 Special marking. Each sack shall be marked with the following precautionary marking: "DO NOT USE FOR MEDICINAL OR EDIBLE PURPOSES"

5.4 Unit loads (Civil agencies only). When specified, sacks of sodium bicarbonate shall be placed on a pallet conforming to PPP-P-0016[illegible] (GSA-FSS) type [illegible], class B or C, grade 2, style and size optional with the supplier. The load shall be secured to the pallet with strapping conforming to QQ-S-781 or PPP-S-760. Edge protectors shall be placed under the straps. A minimum of two straps shall be placed around the top, bottom, and sides of the load and two straps around the top, bottom, and ends of the load. When steel strapping is applied to a load at right angles to each other end in the same plane, it shall be secured from lateral movement and displacement by the application of steel crossties of a size appropriate for the strapping used. Alternatively, the load shall be secured to the pallet by the shrink film process. Thermoplastic films of 6 mil polyethylene, 5.5 mil reinforced polyethylene, or [illegible] mil polyvinyl chloride shall be used. Shrinking of the film shall be in accordance with industry practice.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in soda-acid type fire extinguishers, in chemical munitions and for other purposes where a technical grade of a mildly alkaline material is required. It is not intended for medicinal or other edible purposes.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class of material required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Unit quantity required (see 5.2).
(e) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2.1.1).
(f) Marking document required (see 5.3).
(g) When unit loads are required (see 5.4).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See Section 2 of this specification to obtain extra copies and other documents referenced herein. Price 25 cents each.

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